ponents once they're set into place. The performance requirements in most cases are RAM that matches the processing speed, so that the data is made available as soon as the processor needs it, thus avoiding delays in processing.

Now, you can't just use a RAM stick from your desktop or laptop in an embedded system. Though theoretically the same, there are additional factors that make this impossible:

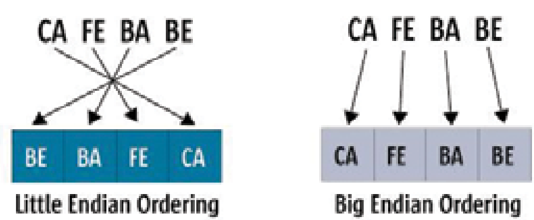
1 Normal desktop/laptop RAM modules are sold separately and can't be soldered into an embedded system's main board directly.

2 Power consumption is always an important issue. There are lower power RAM modules available and they're easily recognisable by the 'L' suffix such as DDR3L. Again, depending on the specific requirements, the RAM might have to be designed for very low voltages or those voltage levels which normal RAM modules are not built for.

3 The method of accessing the data from the memory is equally vital. For example, if the processor accesses data in a 'Big-Endian' style, but the data is stored in RAM in 'Little-Endian' style (Big-endian and little-endian are terms that describe the order in which a sequence of bytes are stored in computer memory.), it will cause some amount of processing to kick in to ensure the correct orientation of data. Hence memory controller and RAM must always be selected/designed with this requirement in mind.

4 If you've seen RAM being sold online, you may have come across the term 'ECC'. It stands for Error Correction Code, which is usually a missing feature on normal RAM modules you can buy from brick-and-mortar stores. The ECC feature ensures that the data stored in the RAM modules is always correct. In situations where the accuracy of data matters a whole lot more than normal, ECC memory is normally required.

We've already mentioned that a graphics card utilises a whole lot of processors and registers, but did you know that one of the cleverest and most important parts of a graphics card design is its RAM? A normal desktop or laptop RAM serves data to the processor in the sequence of requests, which is why companies came up with the idea of multi-channel RAM modules. On the other hand, a graphics card utilises the very clever



Little Endian vs Big Endian memory layout